UNIVERSITY OF WESTMINSTER

INFORMATICS INSTITUTE OF TECHNOLOGY

5COSC020W DATABASE SYSTEMS

Individual Coursework Report

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PART A

QUESTION 1: CRITICAL EVALUATION

Strengths of the AI- Generated EERD

1. Comprehensive includes of major entities.
    - The AI successfully identified most, if not all, of the core business entities described in the brief (e.g., MEMBER, ORDER, PRODUCT, STORE, DELIVERY, PAYMENT and EMPLOYEE). This indicates a strong initial analysis of the natural language requirements.
2. Use of Specialisation (Inheritance).
    - The AI correctly identified that EMPLOYEE is a superclass and that DRIVER and WAREHOUSE_WORKER are subclass of EMPLOYEE
3. Accurate Many to Many resolutions.
    - The AI correctly resolved the critical Many-to-Many relationship STORE_PRODUCT link entity between STORE and PRODUCT

Weaknesses of the AI- Generated EERD

1. Attribute Placement Problems.
    - Some attributes are misplaced or incomplete. For example, licence_no is attached to DRIVER but not clearly linked to VEHICLE.
    - Similarly, capacity_1 needs clarification and should be well defined attribute of VEHICLE.
2. Ambiguous or redundant entities.
    - Concepts such as COLLECTION_ORDER and ORDER-ASSIGNMENT overlap in function and are not well differentiated.
    - This threatens redundancy and confusion in implementation.
3. Weak relationship naming and Multiplicity Justification.
    - Some relations are scarcely annotated (e.g., assigned, uses) with no defined semantic.
    - Not all multiplicities are justified against business rules (e.g., can a driver manage more than one car simultaneously, or only one per delivery?)

Reply

Sathalogithasiva Vyishnavan
Oct 22, 2025 12:40 PM

UOW ID - 20015154

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1. Can a single be offered by multiple stores with different prices and stock levels?
2. Can a basket contain multiple products from the same store, or is it limited to one product per store?
3. Can a single delivery include products from multiple orders, or is it strictly one delivery per order?
4. Are partial payments allowed for large orders, or must the full amount be paid upfront?
5. Can an employee switch roles between warehouse worker and driver, or are roles fixed?

Assumptions Made

1. Add w2001515_StoreProduct associative conceptual entity. This w2001515_StoreProduct entity in the conceptual EERD linking with w2001515_Product and w2001515_Store to find store-specific information. It includes it as an explicit concept Multiplicities are
    - w2001515_Store (1..1)- (1..*) w2001515_StoreProduct
    - w2001515_Product (1..1)- (1..*) w2001515_StoreProduct

2. Hierarchical Modelling of w2001515_PRODUCT specialisation. Perishable products have validity data and drinks must be classified as soft drinks and alcoholic beverages multi-level specialisation was implemented:
    - w2001515_Product is specialised into w2001515_Perishable and w2001515_Non_Perishable under {optional}.
    - w2001515_Perishable is specialised into w2001515_Drink and w2001515_Food under {optional, and}.
    - w2001515_Drink is specialised into w2001515_SoftDrink and w2001515_AlcoholicBeverage under {mandatory, or}.

    This structure supports attribute inheritance and enforces domain-specific constraints.

3. Specialisation of w2001515_Employee into w2001515_Driver and w2001515_Warehouse_Worker under {Optional, Or}. Internal workforce is separated into two different roles with disjoint responsibilities. Disjoint specialization is employed to uniquely classify each employee. This resembles the working structure and also assists data management according to the roles.

QUESTION 2: DECISION LOG

Decisions	Rationale
Add w2001515_StoreProduct associative conceptual entity.	This w2001515_StoreProduct entity in the conceptual EERD linking with w2001515_Product and w2001515_Store to find store-specific information.
Specialisation of w2001515_Employee into w2001515_Driver and w2001515_Warehouse_Worker under {Optional, Or}.	Internal workforce is separated into two different roles with disjoint responsibilities. Disjoint specialization is employed to uniquely classify each employee. This resembles the working structure and also assists data management according to the roles.
Hierarchical Modelling of w2001515_PRODUCT specialisation.	Perishable products have validity data and drinks have to be classified as soft drinks and alcoholic beverages multi-level specialisation was implemented: • w2001515_Product is specialised into w2001515_Perishable and w2001515_Non_Perishable under {optional}. • w2001515_Perishable is specialised into w2001515_Drink and w2001515_Food under {optional, and}. • w2001515_Drink is specialised into w2001515_SoftDrink and w2001515_AlcoholicBeverage under {mandatory, or}. This structure supports attribute inheritance and enforces domain-specific constraints.
w2001515_Vehicle specialisation under {Mandatory, or}	w2001515_TwoWheeler / w2001515_FourWheeler still needed (two-wheelers limited to small collections; four-wheelers for large).
w2001515_Order→ w2001515_WarehouseWorker	orders are allocated to a single warehouse worker to be processed; model accordingly (1:M: Worker processes many orders; each order allocated to one worker at processing time).
Modelling w2001515_Vehicle as a separate entity linked to w2001515_Driver	The drivers operate two-wheelers or four-wheelers and are capacity limited. By modelling w2001515_Vehicle separately, the qualities of vehicle_type and vehicle_capacity can be captured. This enables the delivery logistics and applies the business rule that two-wheelers can complete only small collections.
Inclusion of w2001515_Collection entity for bundled deliveries	According to the case study, several orders may be bundled to be delivered. w2001515_Collection registers what orders are combined, and helps in operating tracking of the logistics

Specialisation of w2001515_Payment into w2001515_CardPaymet and w2001515_MobilePayment under {optional, or}	PAYMENT will be a superclass entity specializing into w2001515_CardPayment and w2001515_MobilePayment. This will be a optional, or specialization as a payment must be one or the other
Prefix all entities and attributes with student ID	As per coursework requirements, all were prefixed with the student ID. This ensures compliance and traceability in assessment.

QUESTION 3: ENTITIES

Entity name	Brief Explanation
w2001515_Member	A registered user who places orders.
w2001515_Store	A registered retailing store that sells groceries via the scheme.
w2001515_Product	A general grocery item offered by stores, identified by a unique code.
w2001515_StoreProduct	An associative entity that links a specific Product to a specific Store, storing its unique price and stock level.
w2001515_NonPerishable	A specialized type of PRODUCT.
w2001515_Perishable	A specialized type of PRODUCT that has validity data.
w2001515_Food	A specialized type of PERISHABLE product.
w2001515_Drink	A specialized type of PERISHABLE product.
w2001515_SoftDrink	A specialized type of DRINK with no alcohol.
w2001515_AlcoholicBeverage	A specialized type of DRINK with alcohol.
w2001515_ShoppingBasket	A named shopping basket created by a member to hold items before ordering.
w2001515_BasketItem	The w2001515_BasketItem entity represents an individual item that a customer has placed into their w2001515_ShoppingBasket.
w2001515_Order	A confirmed order placed by a member based on their basket.
w2001515_OrderItem	Represents a single line item that is part of a completed w2001515_Order.
w2001515_Collection	A bundle of one or more orders assigned to a vehicle for pickup.
w2001515_Delivery	A record of a planned delivery for a single order, assigned to a driver.
w2001515_Employee	An internal workforce member of Foodz@Yourz.
w2001515_Driver	A specialized EMPLOYEE responsible for collecting and delivering orders.
w2001515_WarehouseWorker	A specialized EMPLOYEE responsible for processing orders.

w2001515_Vehicle	A vehicle used to collect orders, either a two-wheeler or four-wheeler.
w2001515_Twowheeler	A specialized VEHICLE with limited volume for small collections.
w2001515_Fourwheeler	A specialized VEHICLE for larger collections.
w2001515_Payment	A record of a single payment taken for an order
w2001515_CardPayment	A specialized type of PAYMENT made by card.
w2001515_MobilePayment	A specialized type of PAYMENT made using a smart phone.

QUESTION 4: SPECIALSATIONS

General entity	Specialised entity	Brief explanation
w2001515_Product	w2001515_NonPerishable	A product with validity data. Specialization is {Optional}
w2001515_Product	w2001515_Perishable	A product with validity data. Specialization is {Optional}
w2001515_Perishable	w2001515_Food	A perishable item that is food. Specialization is {Optional, And}
w2001515_Perishable	w2001515_Drink	A perishable item that is drink. Specialization is {Optional, And}
w2001515_Drink	w2001515_SoftDrink	A drink item that is soft drink. Specialization is {Mandatory, or}
w2001515_Drink	w2001515_AlocholicBeverage	A drink item that is alcoholic beverage. Specialization is {Mandatory, or}
w2001515_Employee	w2001515_Driver	An employee who processes order.

		Specialization is {optional, or}
w2001515_Employee	w2001515_WarehouseWorker	An employee who processes order. Specialization is {optional, or}
w2001515_Payment	w2001515_CardPayment	A payment made by card. Specialization is {Optional, Or}
w2001515_Payment	w2001515_MobilePayment	A payment made by smart phone. Specialization is {Optional, Or}
w2001515_Vehicle	w2001515_Twowheeler	A vehicle for small collections of order. Specialization is {Mandatory, or}
w2001515_Vehicle	w2001515_Fourwheeler	A vehicle for large collections of order. Specialization is {Mandatory, or}

QUESTION 5: RELATIONSHIPS & MULTIPLICITIES

Entity name	Multiplicity	Relationship	Multiplicity	Entity name	Brief justifications for the multiplicity (4 statements for each relationship)
w2001515_Member	0..*	Order from	0..*	w2001515_Store	A single w2001515_Member can place many orders across different w2001515_Stores.
					A w2001515_Member may not have placed any Orders yet from w2001515_Store.
					A single w2001515_Store can receive many Orders from many w2001515_Members.
					A w2001515_Store might not have received any Orders yet from w2001515_Member.
w2001515_Store	1..1	Listed On	1..*	w2001515_StoreProduct	A w2001515_Store must be listed On at least one w2001515_StoreProduct.
					A w2001515_Store can be listed On many w2001515_StoreProduct.
					A w2001515_StoreProduct must be listed by exactly one w2001515_Store.
					A w2001515_StoreProduct cannot be listed by more than one w2001515_Store.
w2001515_Product	1..1	Offer	1..*	w2001515_StoreProduct	A w2001515_Product must be offered as at least one w2001515_StoreProduct.
					A w2001515_Product can be offered as many w2001515_StoreProduct.
					A w2001515_StoreProduct must offer exactly one general w2001515_Product.
					A w2001515_StoreProduct cannot offer more than one general w2001515_Product.

w2001515_OrderItem	1..*	Refer to	1..1	w2001515_Product	Each w2001515_OrderItem must refer to exactly one w2001515_Product.
					A w2001515_OrderItem cannot exist without a related w2001515_Product
					A w2001515_Product may exist even if it hasn't been included in any w2001515_OrderItem.
					A w2001515_Product can be referred to by many w2001515_OrderItems.
w2001515_Order	1..1	Contains	1..1	w2001515_OrderItem	Each w2001515_Order must contain at least one w2001515_OrederItem to be valid.
					A w2001515_Order cannot exist without containing a w2001515_OrderItem.
					A w2001515_OrderItem belongs to exactly one w2001515_Order.
					A w2001515_OrderItem cannot be shared between multiple w2001515_Orders.
w2001515_ShoppingBasket	1..*	Contains	1..1	w2001515_BasketItem	Each w2001515_ShoppingBasket must contain at least one w2001515_BasketItem.
					A w2001515_ShoppingBasket cannot contain many w2001515_BasketItems.
					A w2001515_BasketItem must be contained in at least one w2001515_ShoppingBasket.
					A w2001515_BasketItem can be contained in many different w2001515_ShoppingBaskets.
w2001515_ShoppingBasket	0..*	Contains	0..*	w2001515_Product	A w2001515_ShoppingBasket can contain zero w2001515_Product.
					A w2001515_ShoppingBasket can contain many different w2001515_Products.
					A w2001515_Product can exist without being contained in zero w2001515_ShoppingBasket.
					A w2001515_Product can be contained in many different w2001515_ShoppingBaskets by different members.

w2001515_ShoppingBasket	0..1	Contains	1..1	w2001515_Order	A w2001515_ShoppingBasket can convert to zero w2001515_Order.
					A w2001515_ShoppingBasket can convert to at most one w2001515_Order.
					A w2001515_Order must be converted from exactly one w2001515_ShoppingBasket.
					A w2001515_order cannot be converted from more than one w2001515_ShoppingBasket.
w2001515_Member	1..1	Creates	1..*	w2001515_ShoppingBasket	A w2001515_Member must create at least one w2001515_ShoppingBasket
					A w2001515_Member can create many w2001515_ShoppingBaskets.
					A w2001515_ShoppingBasket must be created by exactly one w2001515_Member.
					A w2001515_ShoppingBasket cannot be creates by many different w2001515_Members.
w2001515_Member	1..1	Places	1..*	w2001515_Order	Each w2001515_Member must have place at least one w2001515_Order after shopping.
					A w2001515_Member can place many w2001515_Orders over time.
					Each w2001515_Order must be placed by exactly one w2001515_Member.
					A w2001515_Order cannot exist without a w2001515_Member who placed it.
w2001515_Collection	0..1	Contains	1..*	w2001515_Order	A w2001515_Collection can contain at least one w2001515_Order.
					A w2001515_Collection can contain many w2001515_Orders.
					A w2001515_Order can be contained in zero w2001515_Collection.
					A w2001515_Order can be contained in at most one w2001515_Collection.
w2001515_Collection	0..*	Result in	0..*	w2001515_Delivery	A w2001515_Collection can result in zero w2001515_Delivery record.

					A w2001515_Collection can result in many w2001515_Delivery records.
					A w2001515_Delivery can result from zero w2001515_Collection record.
					A w2001515_Delivery can result from many w2001515_Collection records.
w2001515_Order	1..*	Belongs	1..1	w2001515_Delivery	Each w2001515_Order belongs to exactly one w2001515_Delivery.
					A w2001515_Order cannot belong to more than one w2001515_Delivery.
					A w2001515_Delivery must belong at least one w2001515_Order to be meaningful.
					Many w2001515_Orders can be belonged into a one w2001515_Delivery.
w2001515_Warehouse Worker	1..1	Processes	1..*	w2001515_Order	Each w2001515_WarehouseWorker must process at least one w2001515_Order.
					A w2001515_WarehouseWorker can handle many w2001515_Orders during their shift.
					A w2001515_Order cannot be processed by many w2001515_WarehouseWorkers simultaneously.
					Every w2001515_Order is processed by exactly one w2001515_WarehouseWorker.
w2001515_Delivery	1..*	Assigned to	1..1	w2001515_Driver	Each w2001515_Delivery must be assigned to exactly one w2001515_Driver.
					A w2001515_Delivery cannot exist without a w2001515_Driver assigned to it.
					A w2001515_Driver can be assigned to at least one w2001515_Delivery.
					A w2001515_Driver can be assigned to many w2001515_Deliveies.

w2001515_Driver	1..*	Collects	1..1	w2001515_Collection	A w2001515_Driver must collect at least one w2001515_Collection.
					A w2001515_Driver can collect many w2001515_Collections.
					A w2001515_Collection must be collected by exactly one w2001515_Driver.
					A w2001515_Collection cannot be collected by more than one w2001515_Driver.
w2001515_Driver	1..1	Drives	1..1	w2001515_Vehicle	Each w2001515_Driver drives exactly one w2001515_Vehicle at a time.
					A w2001515_Driver cannot operate multiple w2001515_Vehicles at a time.
					Each w2001515_Vehicle is assigned to exactly one w2001515_Driver for responsibility tracking.
					A w2001515_Vehicle cannot be shared among multiple w2001515_Drivers at the same time.
w2001515_Order	1..1	Has	1..1	w2001515_Payment	A w2001515_Order must have exactly one w2001515_Payment.
					A w2001515_Order cannot have many w2001515_Payments.
					A w2001515_Payment must be for exactly one w2001515_Order.
					A w2001515_Payment cannot be associated with many w2001515_Orders.

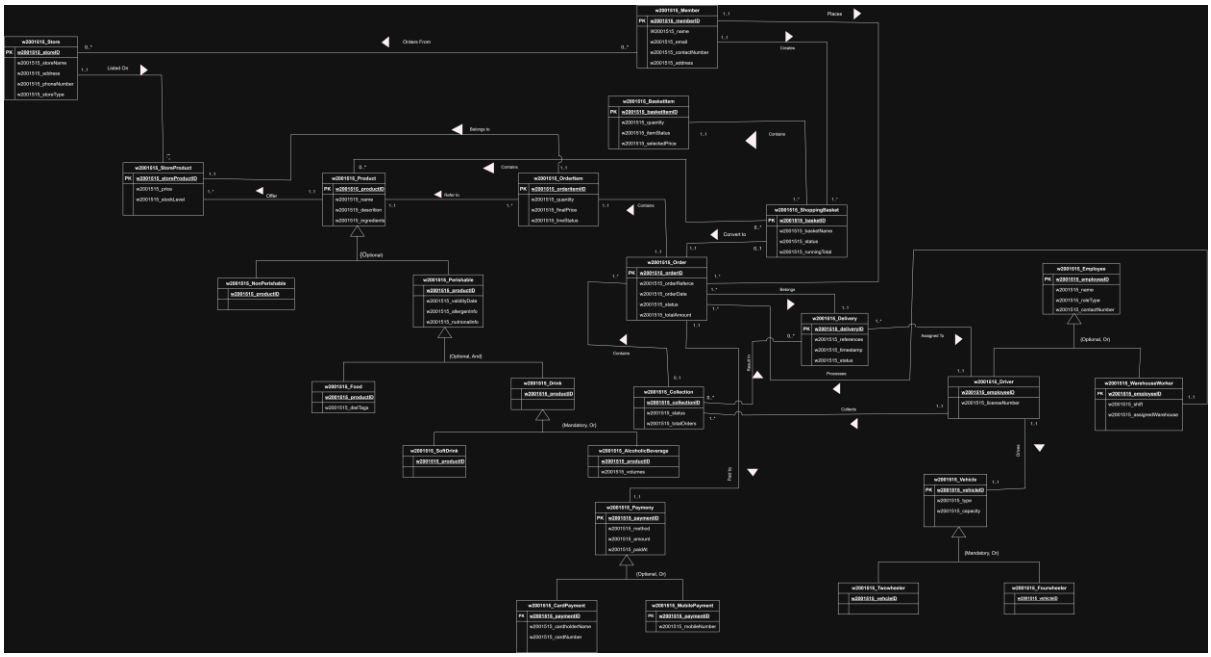
QUESTION 6: ATTRIBUTES & PRIMARY KEYS

Entity name	Attributes for each entity (include PK)	Brief explanation
w2001515_Member	w2001515_memberID (PK), w2001515_name, w2001515_email, w2001515_contactNumber, w2001515_address	Core details of the registered member
w2001515_Store	w2001515_storeID (PK), w2001515_storeName, w2001515_address,	Core details of the participating store

	w2001515_phoneNumber, w2001515_storeType	
w2001515_Product	w2001515_productID (PK), w2001515_name, w2001515_descrition, w2001515_ingredients	Detail of the product
w2001515_StoreProduct	w2001515_storeProductID (PK), w2001515_price, w2001515_stockLevel	An associative entity that links a specific Product to a specific Store, storing its unique price and stock level.
w2001515_NonPreishable		Inherits from PRODUCT
w2001515_Perishable	w2001515_validityDate, w2001515_allergenInfo, w2001515_nutrionalInfo	Inherits from PRODUCT; this perishable has validity data.
w2001515_Food	w2001515_dietTags	Inherits from PERISHABLE; this food has validity data.
w2001515_Drink		Inherits from PERISHABLE
w2001515_Softdrink		Inherits from DRINK
w2001515_AlcoholicBeverage	w2001515_volumes	Inherits from DRINK; this alcoholic beverage has validity data.
w2001515_ShoppingBasket	w2001515_basketID (PK), w2001515_basketName, w2001515_status, w2001515_runningTotal	Details of a member's shopping basket.
w2001515_BasketItem	w2001515_basketItemID (PK), w2001515_quantity, w2001515_itemStatus, w2001515_selectedPrice	The w2001515_BasketItem entity represents an individual item that a customer has placed into their w2001515_ShoppingBasket.
w2001515_Order	w2001515_orderID (PK), w2001515_orderReferce, w2001515_orderDate, w2001515_status, w2001515_totalAmount	Represents a single line item that is part of a completed w2001515_Order.
w2001515_OrderItem	w2001515_orderItemID (PK), w2001515_quantity, w2001515_finalPrice, w2001515_lineStatus	The w2001515_OrderItem entity
w2001515_Collection	w2001515_collectionID (PK), w2001515_status, w2001515_totalOrders	Details of a delivery collection,
w2001515_Delivery	w2001515_deliveryID (PK), w2001515_references,	Details of the delivery of the product; it holds reference, timestamp and status

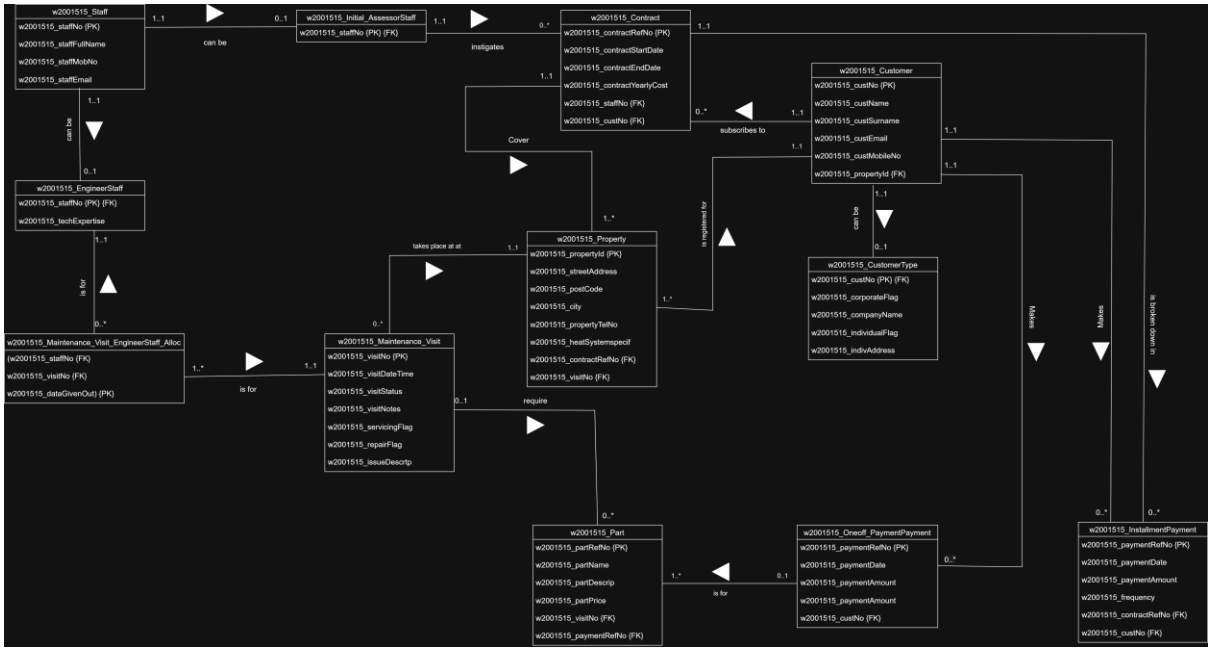
	w2001515_timestamp, w2001515_status	
w2001515_Employee	w2001515_employeeID (PK), w2001515_name, w2001515_roleType, w2001515_contactNumber	General employee details
w2001515_Driver	w2001515_licenseNumber	Inherits from EMPLOYEE; its store driver's license
w2001515_WarehouseWorker	w2001515_shift, w2001515_assignedWarehouse	Inherits from EMPLOYEE; its store worker's shift and the assigned warehouse
w2001515_Vehicle	w2001515_vehicleID (PK), w2001515_type, w2001515_capacity	General vehicle details
w2001515_Twowheeler		Inherits from VEHICLE
w2001515_Fourwheeler		Inherits from VEHICLE
w2001515_Payment	w2001515_paymentID (PK), w2001515_method, w2001515_amount, w2001515_paidAt	General payment details, linked to an order
w2001515_CardPayment	w2001515_cardholderName, w2001515_cardNumber	Inherits from PAYMENT; it stores cardholder name and card number
w2001515_MobilePayment	w2001515_mobileNumber	Inherits from PAYMENT; it store mobile number.

QUESTION 7: QUALITY & NOTATIONS OF CONCEPTUAL EERD



PART B

QUESTION 8: MAPPING CONCEPTUAL EERD TO LOGICAL ERD



QUESTION 9: MAPPING LOG

1. Solved the specialization.
    - Solved the specialization of staff entity {Optional, Or}
        - The primary key in the staff table is used as the primary key of child table (w2001515_Engineer, w2001515_Initial_Assessor) which are former specialized entities.
    - Solved the specialization of Maintenance_Visit {mandatory, and}
        - As the mapping rules of {mandatory, and} specialization Service and Repair entity attributes merged with a w2001515_Maintenance_Visit entity.
    - Solved the specialization of Payment {mandatory, or}
        - As the mapping rules of {mandatory, or} all entities in the specialization are create as separate entities. The general entity is erased (example: - the specialized entities w2001515_Oneoff_PaymentPayment and w2001515_IntallmentPayment are created as separate entity tables and the Payment table is erased) all the attributes of the general entity (Payment) are added to the new specialized entity tables. The relationships associated with the general entity are doubled and applied to each specialized entity table. Relationships specific to the specialized entity remain the same.
    - Solved the specialization of Customer {optional, and}
        - As a mapping rules the customer table is taken as the parent table (w2001515_Customer) and both specialization entity tables merged and created a child entity (w2001515_CustomerType). The primary key of parent table w2001515_custNo is both primary and foreign key of the child table.
2. Association entities
    - The Association entity w2001515_Maintenance_Visit_EngineerStaff_Alloc resolves the Mant-to-Many relationships between w2001515_EngineerStaff and the w2001515_Maintenance_Visit. It includes foreign keys {FK} w2001515_staffNo and w2001515_visitNo as a composite primary key {PK} to uniquely identify each allocation.
3. Solved one to many relationship
    - According to the one-to-many relationship created 2 entity tables based on the original 2 entities. Considering the multiplicities the “One” side is chosen as parent table and “many” sides is chosen as child table. And

the PK of parent is created as a FK in child table For example :- (w2001515_Maintenance_Visit and w2001515_Part, w2001515_Oneoff_PaymentPayment and w2001515_Part, w2001515_Customer and w2001515_Oneoff_PaymentPayment, w2001515_Customer and w2001515_InstallmentPayment, w2001515_Initial_AssessorStaff and w2001515_Contract, w2001515_Contract and w2001515_Property, w2001515_Property and w2001515_Customer, w2001515_Contract and w2001515_Customer, w2001515_Contract and w2001515_InstallmentPayment).